Exercise 06 - Runtimes

For each function listed below, construct its runtime table and calculate its runtime function for its worst-case scenario. Afterward, state and prove the theta-notation of each function by providing the values of c_1, c_2, and n_1 from the definition of theta-notation.

Functions:

```
size_t A(size_t n)
{
  size_t c = 0;

  for(size_t i = 0;i < n;i += 1)
  {
    if(n % i != 0)
    {
      c += 1;
    }
  }
  return c;
}
```

```
bool B(const Array<size_t>& data)
{
  long c;

  for(size_t i = 0;i < data.size();i += 1)
  {
    if(data[i] % 2 == 0)
    {
      c += 1;
    }
  }
  return data.size() - c;
}
```

```
bool C(const string& str)
{
  for(int i = 0;i < str.size();i += 1)
  {
    for(int j = i + 1;j < str.size();j += 1)
    {
      if(str[i] == str[j])
      {
        return true;
      }
    }
  }
  return false;
}
```

```
bool D(Array<bool>& data)
{
  size_t i= 0;
  size_t j = data.size() - 1;
  size_t c = 0;
  while(i < j && data[i] == data[j])
  {
    i += 1;
    j -= 1;
    c += 1;
  }
  c >= 2;
}
```

```
bool E(string str)
{
  bool found = false;

  for(size_t i = 1;i < str.size();i += 1)
  {
    if(found)
    {
      if(data[i-1] == data[i])
      {
        return true;
      }
      else
      {
        found = false;
      }
    }
    else if(data[i-1] == data[i])
    {
      found = true;
    }
  }
  return false;
}
```